

26021 Kemi (Polyteknisk grundlag)/26021 Chemistry (Polytechnic Foundation) non- random version

Der er kun et korrekt svar for hvert spørgsmål.
Du kan kun vælge ét svar per spørgsmål
Alle spørgsmål indgår med samme vægt i bedømmelsen.

There is only one correct answer for each question.
You can only select one answer per question.
All questions have equal weight in the evaluation.

Hvad er antallet af neutroner i ^{238}U ?

What is the number of neutrons in ^{238}U ?

Choose one answer

☐ 92

☒ 146

☐ 238

☐ 0

☐ 119

Hvad er den kemiske formel for ammoniumnitrat?

What is the chemical formula for ammonium nitrate?

Choose one answer

☐ $\text{Am}(\text{NO}_3)_4$

☐ NH_4NO_3

☐ $(\text{NH}_4)_3\text{N}$

☒ NH_4NO_2

☐ $\text{Am}(\text{NO}_3)_3$

Hvad er oxidationstallet for Cr i $\text{Cr}_2\text{O}_7^{2-}$?

What is the oxidation state for Cr in $\text{Cr}_2\text{O}_7^{2-}$?

Choose one answer

☐ 0

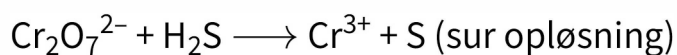
☐ 3

☐ 7

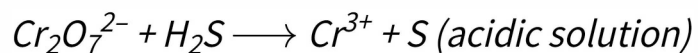
☒ 6

☐ 12

Når den følgende reaktion er afstemt med det numerisk mindst mulige set af heltal, hvad er summen af koefficienterne (glem ikke koefficienter af 1):



When the following reaction is balanced with the smallest possible set of integers, what is the sum of all the coefficients (do not forget coefficients of one)



Choose one answer

☐ 29

☐ 13

☒ 24

☐ 19

☐ 7

Hvad er massen af 1 mmol (millimol) ammoniumhydrogenphosphat?

What is the mass of 1 mmol (millimol) ammonium hydrogenphosphate?

Choose one answer

- ☐ 115 g
- ☐ 132 g
- ☐ 1.15 g
- ☒ 0.132 g
- ☐ $6.02 \cdot 10^{-20}$ g

Hvilken af følgende har den største masse?

Which of the following have the largest mass?

Choose one answer

☒ 1 mol SF₆(g)

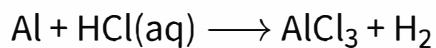
☐ 1 mol Fe(s)

☐ 1 mol CO₂(g)

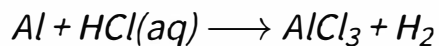
☐ De har alle samme masse [*They all have the same mass*]

☐ 1 mol CH₃CH₂OH(l)

Hvad er massen af dihydrogen der dannes når aluminium (25 g) reagerer med saltsyre ifølge denne (ikke afstemte) reaktion:



What is the mass of dihydrogen formed when aluminium (25 g) reacts with hydrochloric acid according the (unbalanced) reaction:



Choose one answer

☒ 2.8 g

☐ 0.41 g

☐ 1.9 g

☐ 0.93 g

☐ 1.2 g

QUESTION 8

En koncentrationscelle er konstrueret ved at placere identiske kobberelektroder i to kobber(II)-opløsninger. De to opløsninger er 1.0 M og 2.0 mM i Cu^{2+} -koncentration. Udregn den elektromotoriske kraft for den spontant forløbende process i koncentrationscellen, E_{cell} .

A concentration cell is constructed by placing identical copper electrodes in two copper(II) solutions. The two solutions are 1.0 M and 2 mM in Cu^{2+} concentration. Calculate the electromotive force of the spontaneous process in the concentration cell, E_{cell} .

Choose one answer

- ☐ -0.080 V
- ☒ +0.080 V
- ☐ +0.020 V
- ☐ -0.34 V
- ☐ 0.34 V

Hvilken af følgende er den stærkeste syre?

Which of the following is the strongest acid?

Choose one answer

☒ $\text{H}_2\text{SO}_4(\text{aq})$

☐ $\text{H}_2\text{O}(\text{l})$

☐ $\text{H}_3\text{PO}_4(\text{aq})$

☐ $\text{H}_2\text{PO}_4^-(\text{aq})$

☐ $\text{HPO}_4^{2-}(\text{aq})$

Betragt en given kemisk reaktion og dets ligevægtskonstant, K . Hvilket af følgende udsagn er korrekt?

Consider a given chemical reaction and its equilibrium constant, K . Which of the following statements is correct?

Choose one answer

- ☒ K ændres hvis temperaturen ændres [*K changes if the temperature changes*]
- ☐ K forbliver den samme hvis temperaturen ændres [*K remains the same if the temperature changes*]
- ☐ K formindskes hvis koncentrationen af en reaktant formindskes [*K decreases if the concentration of a reactant is decreased*]
- ☐ K forøges hvis koncentrationen af en af produkterne forøges [*K increases if the concentration of one of the products is increased*]
- ☐ K ændres hvis en katalysator bliver tilført [*K changes by if a catalyst is added*]

Hvilket volumen NaOH(aq) (0.5 M) vil du have brug for, for at omdanne en 500 mL-opløsning af H_3PO_4 (0.25 M) til en opløsning af natriumphosphat?

What volume of NaOH(aq) (0.50 M) would you need to convert a 500 mL-solution of H_3PO_4 (0.25 M) into a solution of sodium phosphate?

Choose one answer

- ☒ 0.75 L
- ☐ $2.5 \cdot 10^2$ mL
- ☐ $1.4 \cdot 10^2$ mL
- ☐ 83 mL
- ☐ 1.5 L

Betragt en opløsning som er 0.10 M i CH_3COOH og 0.20 M i NaCH_3COO . Hvilket af følgende udsagn er korrekt?

Consider a solution which is 0.10 M in CH_3COOH and 0.20 M in NaCH_3COO . Which of the following statements is true?

Choose one answer

- ☐ Hvis en lille mængde OH^- tilsættes, formindskes pH ganske lidt. [*If a small amount of OH^- is added, the pH decreases very slightly*]
- ☐ Hvis natriumhydroxid tilsættes vil OH^- reagere med CH_3COO^- . [*If sodium hydroxide is added, OH^- reacts with CH_3COO^-*]
- ☒ Hvis en lille mængde HCl(aq) tilsættes, formindskes pH meget lidt. [*If a small amount of HCl(aq) is added, the pH decreases very slightly*]
- ☐ Hvis saltsyre tilsættes reagerer H^+ med CH_3COOH . [*If hydrochloric acid is added, the H^+ ions react with CH_3COOH*]
- ☐ Hvis der tilsættes mere CH_3COOH , vil pH forøges. [*If more CH_3COOH is added, the pH increases*]

Tre flasker (2 L) fyldes respektivt med H_2 , O_2 og He, ved $T = 25\text{ }^\circ\text{C}$ and $P = 1\text{ bar}$. Hvilket af følgende udsagn er korrekt? Gasserne kan betragtes som værende ideale.

Three flasks (2 L) are filled with H_2 , O_2 , and He, respectively, at $T = 25\text{ }^\circ\text{C}$ and $P = 1\text{ bar}$. Which of the following statements is correct? The gases can be considered as ideal.

Choose one answer

- ☐ Molekylernes hastighed er identisk i alle flaskerne. [*The velocity of the gas molecules is identical in each flask*]
- ☒ Antallet af O_2 -molekyler i den ene flaske er identisk med antallet af He-atomer i den anden. [*The number of O_2 molecules in one flask is identical to the number of He atoms in the other flask*]
- ☐ Densiteten af gasserne er den samme. [*The density of each gas is the same*]
- ☐ Der er dobbelt så mange H_2 -molekyler som der er He-atomer. [*There are twice as many H_2 molecules as there are He atoms*]
- ☐ Ingen af disse er korrekte. [*None of these are correct*]

QUESTION 14

De følgende opløselighedsprodukter for chromater (CrO_4^{2-} -salte) er blevet bestemt (ved 25 °C):

	K_{sp}
Ag_2CrO_4	$9.0 \cdot 10^{-12} \text{ M}^3$
BaCrO_4	$2.0 \cdot 10^{-10} \text{ M}^2$
PbCrO_4	$1.8 \cdot 10^{-14} \text{ M}^2$

Chromatet der er **mest** opløseligt (i mol/L) ved 25 °C er:

The following solubility products for three chromates (CrO_4^{2-} salts) have been determined (at 25 °C):

	K_{sp}
Ag_2CrO_4	$9.0 \cdot 10^{-12} \text{ M}^3$
BaCrO_4	$2.0 \cdot 10^{-10} \text{ M}^2$
PbCrO_4	$1.8 \cdot 10^{-14} \text{ M}^2$

*The chromate that is the **most** soluble (in mol/L) at 25 °C is:*

Choose one answer

- ☐ BaCrO_4
- ☒ Ag_2CrO_4
- ☐ Alle er fuldt opløselige i vand. [*All are fully soluble in water*]
- ☐ PbCrO_4
- ☐ Umuligt at bestemme uden yderligere information. [*Impossible to determine without additional information*]

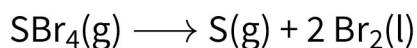
Atomet med elektronkonfigurationen $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^5$ vil være i:

The atom with the electron configuration $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^5$ would be in:

Choose one answer

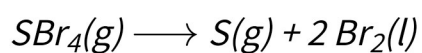
- ☐ 6. hovedgruppe, 5. periode [*Main group 6, period 5*]
- ☒ 7. hovedgruppe, 4. periode [*Main group 7, period 4*]
- ☐ 4. hovedgruppe, 4. periode [*Main group 4, period 4*]
- ☐ 4. hovedgruppe, 7. periode [*Main group 4, period 7*]
- ☐ 6. hovedgruppe, 5. periode [*Main group 6, period 5*]

Betragt følgende reaktion:



ved $T = 25\text{ }^\circ\text{C}$, $\Delta H^\circ = +115\text{ kJ mol}^{-1}$ og $\Delta S^\circ = +125\text{ J K}^{-1}\text{ mol}^{-1}$. Udregn ΔG° for reaktionen ved $25\text{ }^\circ\text{C}$.

Consider the following reaction:



at $T = 25\text{ }^\circ\text{C}$, $\Delta H^\circ = +115\text{ kJ mol}^{-1}$ and $\Delta S^\circ = +125\text{ J K}^{-1}\text{ mol}^{-1}$. Calculate ΔG° for the reaction at $25\text{ }^\circ\text{C}$.

Choose one answer

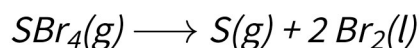
- ☐ -56.7 kJ mol^{-1}
- ☒ $+77.8\text{ kJ mol}^{-1}$
- ☐ -86.2 kJ mol^{-1}
- ☐ $+152\text{ kJ mol}^{-1}$
- ☐ $+112\text{ kJ mol}^{-1}$

Betragt igen reaktionen:



ved $T = 25\text{ }^\circ\text{C}$, $\Delta H^\circ = +115\text{ kJ mol}^{-1}$ og $\Delta S^\circ = +125\text{ J K}^{-1}\text{ mol}^{-1}$ (som begge kan antages som værende temperaturuafhængige). Denne reaktion er:

Consider again the reaction:



at $T = 25\text{ }^\circ\text{C}$, $\Delta H^\circ = +115\text{ kJ mol}^{-1}$ and $\Delta S^\circ = +125\text{ J K}^{-1}\text{ mol}^{-1}$ (both of which can be assumed independent of temperature). This reaction is:

Choose one answer

- ☒ Spontan ved temperaturer over ca. 900 K. [*Spontaneous at temperatures above ca. 900 K*]
- ☐ Spontan ved alle temperaturer. [*Spontaneous at all temperatures*]
- ☐ Spontan ved temperaturer under ca. 900 K. [*Spontaneous at temperatures below ca. 900 K*]
- ☐ Kun spontan ved 25 °C. [*Only spontaneous at 25 °C*]
- ☐ Ikke-spontan ved alle temperaturer. [*Non-spontaneous at all temperatures*]

De fleste kemiske reaktioner forløber hurtigere ved højere temperaturer end ved lavere temperaturer. Dette skyldes:

- (I) en forøgelse i reaktionens aktiveringsenergi, E_a , med stigende temperatur.
- (II) en forøgelse af reaktionens hastighedskonstant, k , med stigende temperatur.
- (III) en forøgelse i molekylers gennemsnitlige hastighed med stigende temperatur.

The majority of chemical reactions are more rapid at higher temperatures than at lower temperatures. This observation is due to:

- (I) an increase in the activation energy, E_a , of the reaction with increasing temperature.*
- (II) an increase in the rate constant, k , with increasing temperature.*
- (III) an increase in the average velocity of the molecules with increasing temperature.*

Choose one answer

- ☐ Kun (I). [*Only (I)*]
- ☐ Kun (II). [*Only (II)*]
- ☐ Både (I), (II) og (III). [*All of (I), (II), and (III)*]
- ☒ Kun (II) og (III). [*Only (II) and (III)*]
- ☐ Kun (I) og (II). [*Only (I) and (II)*]

Hvilke(t) af følgende fire molekyler er polært: PH_3 , H_2O , HCl , SO_3 ?

Which of the following more molecules is/are polar: PH_3 , H_2O , HCl , SO_3 ?

Choose one answer

- ☐ Kun HCl og H_2O . [*Only HCl and H_2O*]
- ☐ Kun HCl . [*Only HCl*]
- ☒ Alle undtagen SO_3 . [*All except SO_3*]
- ☐ Ingen af dem. [*None of them*]
- ☐ Allesammen. [*All of them*]

QUESTION 20

For grundstofferne i 2. hovedgruppe, forøges alle de følgende når man går ned igennem gruppen, pånær:

For the main group 2 elements, all of the following increase with going down the group except:

Choose one answer

- ☐ Atomets volumen. [*The volume of the atom*]
- ☒ Den første ioniseringsenergi. [*The first ionization energy*]
- ☐ Den molare masse. [*The molar mass*]
- ☐ Ionradius. [*The ionic radius*]
- ☐ Atomradius. [*The atomic radius*]

